

Lab Guide 3.1 – Process management

Target	Related theory	Resources
Process concept	Ch3: Process Creation	Use Ubuntu 16/18 image
Observe the process in the system.	Ch3: Process Creation	Use Windows 10 Use Cocalc online

Update

2021.Jun.2	Add progress tracking.
2023.Mar.10	Add Cocalc part (2B)

Objective: Understand the theory involved. Know how to compile sample code and execute it. Know how to observe processes inside Windows and Ubuntu systems. Apply the commands introduced to complete the exercise.

Student reviews: Questions and answers about theoretical issues. Practical skills. Exercise.

Submission requirements: .c source files and .out executable files for the “examples” and exercises at the end of the tutorial. Screenshot of progress information.

Preferences

[1] Abraham Silberschatz, Peter B. Galvin, Greg Gagne, [2018], Operating System Concepts, 10th edition, John Wiley & Sons, New Jersey.

Programming Problems of Chapter 3.

[2] Greg Gagne , [2019], GitHub OS-BOOK OSC10e, Westminster College, United States

Access <https://github.com/greggagne/osc10e> in September 2019.

[3] Linux manual page, [2021], on getpid(2)

Access <https://man7.org/linux/man-pages/man2/getpid.2.html>

1. Process management in GUI.

?	- What is the process?	Ch3 p.106	Slide Ch3.4
	- When is the process born? And when does it end?	Ch3 p.116	Slide Ch3.20
	- What is the difference between Process and Program?	Ch3 p.121	Slide Ch3.25

Requirements 1: Open the Windows Task Manager application by pressing Ctrl + Alt + Del and selecting Task Manager.

Observe the Process tab:

- Explain how Apps and Background Process are different?
- Which application is using the most CPU, which application is being allocated the most RAM.
- Operate on several applications (for example: browser) and observe information changes in the Process tab.
- Sort applications by CPU Usage from high to low.

Task Manager

File Options View

Processes Performance App history Startup Users Details Services

Name	Status	42% CPU	81% Memory	1% Disk	0% Network	0% GPU	GPU engi...	Power us...	Power us...
Apps (6)									
Brave Browser (10)		0.2%	748.2 MB	0.1 MB/s	0 Mbps	0%		Very low	Very low
Microsoft Word (2)		0%	109.0 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Task Manager		1.6%	33.9 MB	0 MB/s	0 Mbps	0%		Very low	Very low
VirtualBox Manager		34.1%	43.5 MB	0.1 MB/s	0 Mbps	0%		Very high	Moderate
VirtualBox Manager		0%	2.0 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Windows Explorer		0.7%	24.7 MB	0.1 MB/s	0 Mbps	0%		Very low	Very low
Background processes (108)									
64-bit Synaptics Pointing Enhanc...		0%	0.4 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Antimalware Service Executable		0.4%	87.1 MB	0.1 MB/s	0 Mbps	0%		Very low	Very low
Application Frame Host		0%	3.9 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Bluetooth OBEX Service		0%	0.4 MB	0 MB/s	0 Mbps	0%		Very low	Very low

^ Fewer details

End task

Figure 1. Process tab of the Task Manager application.

Observe the Details tab

Task Manager

File Options View

Processes Performance App history Startup Users Details Services

Name	PID	Status	User name	CPU	Memor...	UAC virtua...
System Idle Process	0	Running	SYSTEM	87	8 K	
audiodg.exe	10220	Running	LOCAL SE...	06	8,280 K	Not allowed
Taskmgr.exe	15160	Running	sony	02	24,000 K	Not allowed
dwm.exe	14004	Running	DWM-4	01	47,372 K	Disabled
svchost.exe	848	Running	SYSTEM	01	10,836 K	Not allowed
VirtualBox.exe	9192	Running	sony	00	54,676 K	Disabled
svchost.exe	4724	Running	LOCAL SE...	00	33,880 K	Not allowed
explorer.exe	5232	Running	sony	00	22,860 K	Disabled
csrss.exe	13652	Running	SYSTEM	00	1,184 K	Not allowed
System	4	Running	SYSTEM	00	20 K	
svchost.exe	1772	Running	LOCAL SE...	00	7,428 K	Not allowed
WINWORD.EXE	3848	Running	sony	00	193,720 K	Disabled
nativeproxy.exe	12204	Running	sony	00	336 K	Disabled
VESMgrSub.exe	10980	Running	SYSTEM	00	1,248 K	Not allowed

^ Fewer details

End task

Figure 2. Details tab of the Task Manager application.

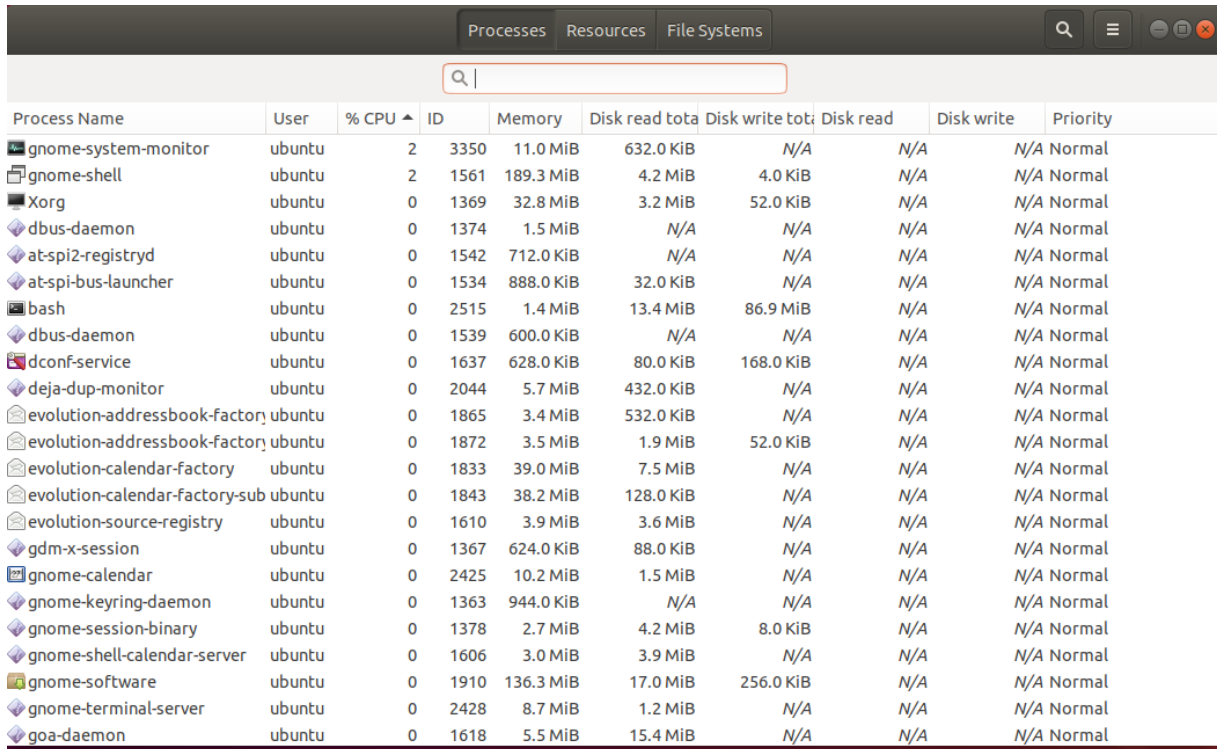
- Process A (or some other process), what is its ID, who started it and what is its status?

- The "System Idle Process" process is a special process that always uses the "rest" of the CPU.
Why is there a need for such a process inside the Operating System?

Requirement 2:

- Perform the operations as in Requirement 1.
- What are the similarities and differences between Task Manager and System Monitor / Processes?

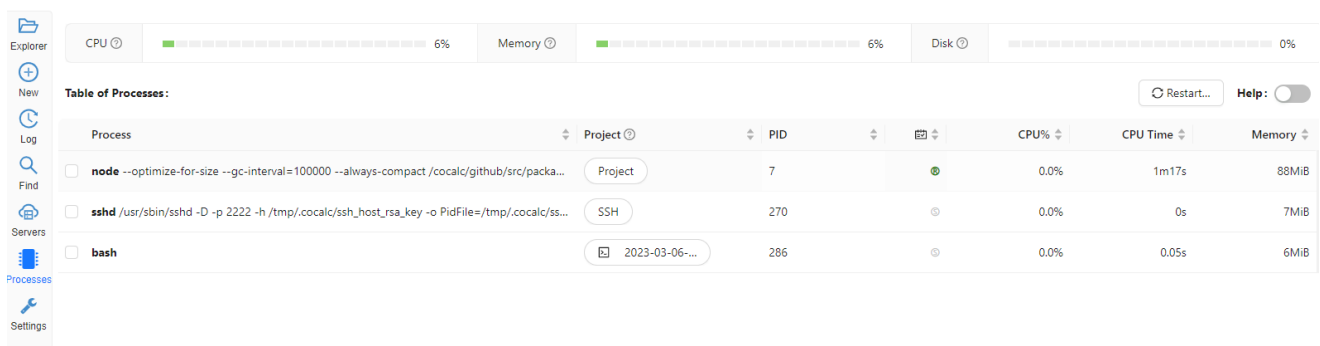
- If using Ubuntu, open the System Monitor application.



Process Name	User	% CPU	ID	Memory	Disk read tota	Disk write tota	Disk read	Disk write	Priority
gnome-system-monitor	ubuntu	2	3350	11.0 MiB	632.0 KiB	N/A	N/A	N/A	Normal
gnome-shell	ubuntu	2	1561	189.3 MiB	4.2 MiB	4.0 KiB	N/A	N/A	Normal
Xorg	ubuntu	0	1369	32.8 MiB	3.2 MiB	52.0 KiB	N/A	N/A	Normal
dbus-daemon	ubuntu	0	1374	1.5 MiB	N/A	N/A	N/A	N/A	Normal
at-spi2-registryd	ubuntu	0	1542	712.0 KiB	N/A	N/A	N/A	N/A	Normal
at-spi-bus-launcher	ubuntu	0	1534	888.0 KiB	32.0 KiB	N/A	N/A	N/A	Normal
bash	ubuntu	0	2515	1.4 MiB	13.4 MiB	86.9 MiB	N/A	N/A	Normal
dbus-daemon	ubuntu	0	1539	600.0 KiB	N/A	N/A	N/A	N/A	Normal
dconf-service	ubuntu	0	1637	628.0 KiB	80.0 KiB	168.0 KiB	N/A	N/A	Normal
deja-dup-monitor	ubuntu	0	2044	5.7 MiB	432.0 KiB	N/A	N/A	N/A	Normal
evolution-addressbook-factory	ubuntu	0	1865	3.4 MiB	532.0 KiB	N/A	N/A	N/A	Normal
evolution-addressbook-factory	ubuntu	0	1872	3.5 MiB	1.9 MiB	52.0 KiB	N/A	N/A	Normal
evolution-calendar-factory	ubuntu	0	1833	39.0 MiB	7.5 MiB	N/A	N/A	N/A	Normal
evolution-calendar-factory-sub	ubuntu	0	1843	38.2 MiB	128.0 KiB	N/A	N/A	N/A	Normal
evolution-source-registry	ubuntu	0	1610	3.9 MiB	3.6 MiB	N/A	N/A	N/A	Normal
gdm-x-session	ubuntu	0	1367	624.0 KiB	88.0 KiB	N/A	N/A	N/A	Normal
gnome-calendar	ubuntu	0	2425	10.2 MiB	1.5 MiB	N/A	N/A	N/A	Normal
gnome-keyring-daemon	ubuntu	0	1363	944.0 KiB	N/A	N/A	N/A	N/A	Normal
gnome-session-binary	ubuntu	0	1378	2.7 MiB	4.2 MiB	8.0 KiB	N/A	N/A	Normal
gnome-shell-calendar-server	ubuntu	0	1606	3.0 MiB	3.9 MiB	N/A	N/A	N/A	Normal
gnome-software	ubuntu	0	1910	136.3 MiB	17.0 MiB	256.0 KiB	N/A	N/A	Normal
gnome-terminal-server	ubuntu	0	2428	8.7 MiB	1.2 MiB	N/A	N/A	N/A	Normal
goa-daemon	ubuntu	0	1618	5.5 MiB	15.4 MiB	N/A	N/A	N/A	Normal

Figure 3. The gnome-system-monitor application.

- If using Cocalc, open the Processes utility.



Process	Project	PID	CPU%	CPU Time	Memory
node --optimize-for-size --gc-interval=100000 --always-compact /cocalc/github/src/packa...	Project	7	0.0%	1m17s	88MiB
sshd /usr/sbin/sshd -D -p 2222 -h /tmp/cocalc/ssh_host_rsa_key -o PidFile=/tmp/cocalc/ss...	SSH	270	0.0%	0s	7MiB
bash	2023-03-06-...	286	0.0%	0.05s	6MiB

- Use the htop command in Terminal

```

1  [|||||] 10.0% Tasks: 174, 826 thr: 1 running
2  [|||||] 11.2% Load average: 8.30 5.71 3.55
3  [|||||] 9.9% Uptime: 01:37:33
4  [|||||] 12.6%
Mem[|||||] 6.44G/7.68G
Swp[|||||] 268K/2.00G

PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
10942 mtravisro 20 0 1044M 351M 85192 S 11.2 4.5 1:52.09 /opt/google/chrome/chrome --type=rend
10885 mtravisro 20 0 1035M 337M 105M S 4.6 4.3 1:24.77 /opt/google/chrome/chrome --type=rend
10930 mtravisro 20 0 1114M 381M 281M S 3.3 4.9 2:35.76 /opt/google/chrome/chrome --type=rend
3317 mtravisro 20 0 1245M 19032 13560 S 3.3 0.2 0:27.52 /usr/bin/pulseaudio --start --log-tar
4488 mtravisro 20 0 1492M 377M 149M S 3.3 4.8 10:45.90 /opt/google/chrome/chrome
2814 root 20 0 384M 69212 53596 S 3.3 0.9 2:35.01 /usr/lib/xorg/Xorg -dpi $DPI :0 -seat
11528 mtravisro 20 0 1114M 381M 281M S 2.6 4.9 0:02.11 /opt/google/chrome/chrome --type=rend
11795 mtravisro 20 0 60492 4408 3036 R 2.6 0.1 0:09.68 htop
11454 mtravisro 20 0 1101M 295M 96928 S 2.0 3.8 0:52.26 /opt/google/chrome/chrome --type=rend
3429 mtravisro 20 0 1245M 19032 13560 S 2.0 0.2 0:17.04 /usr/bin/pulseaudio --start --log-tar
12165 mtravisro 20 0 1044M 351M 85192 S 2.0 4.5 0:00.62 /opt/google/chrome/chrome --type=rend
4508 mtravisro 20 0 1492M 377M 149M S 2.0 4.8 4:21.23 /opt/google/chrome/chrome
11534 mtravisro 20 0 1035M 337M 105M S 2.0 4.3 0:03.68 /opt/google/chrome/chrome --type=rend
5482 mtravisro 20 0 1132M 402M 112M S 2.0 5.1 8:26.27 /opt/google/chrome/chrome --type=rend
11398 mtravisro 20 0 758M 125M 80244 S 2.0 1.6 0:08.86 /opt/google/chrome/chrome --type=rend
9884 mtravisro 20 0 1012M 53444 42948 S 1.3 0.7 0:03.60 /opt/google/chrome/chrome --type=util
11535 mtravisro 20 0 1035M 337M 105M S 1.3 4.3 0:03.99 /opt/google/chrome/chrome --type=rend
12170 mtravisro 20 0 1044M 351M 85192 S 1.3 4.5 0:00.63 /opt/google/chrome/chrome --type=rend

F1Help F2Setup F3Search F4Filter F5Free F6SortBy F7Nice F8nice F9Kill F10Quit

```

2. Process management in CLI

Requirement 3: Execute process management commands in Ubuntu Terminal.

- List progress:

```
> ps [options]
```

- Options:

- -a: displays linked user processes
- -e: show all progress
- -f: show PID of parent process and start time
- -l: similar to -f

- Cancel process:

```
> kill [-signal | -s signal] pid
```

Often combined with the ps command to get the process ID

- Signals:

- 2: equivalent to CTRL + C
- 9: force the end
- 15: default – smooth ending
- 19: pause
- Kill – l: lists all signals

- For example:

```
kill -9 11234 (kill process with pid = 11234)
kill -s 9 11234
```

3. Process Management calls

Requirement 4: Use functions to get process information. The library “unistd.h” contains system calls to create processes fork() and view their IDs getpid() and view the ID of the parent process getppid().

Instruct: Download the file LAB_IntroOS/LAB_3/task1_4.c, then compile and execute to know the ID of this process along with the ID of the parent process.

- Line 5 is the library containing the calls in lines 7 and 8.
- Line 7 contains a call that returns the ID of the running process.
- Line 8 contains a call that returns the ID of the parent process.

```
5    #include <unistd.h>
7    printf("Current process ID: %d\n", getpid());
8    printf("Parent process ID: %d\n", getppid())
```

?	- What is the process ID? - What is the ID of the parent process? - Open System Monitor and find which processes have the IDs mentioned above.	Ch3 p.107	Slide Ch3.10
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Requirement 5: add the sleep(5) command after line 7 and before line 8. Compile and run the program again, at the same time switch to the "System Monitor" utility to search for process ID information, observe changes when The process ends after 5 seconds.

Note: parameter 5 in the sleep command means the program pauses for 5 seconds. SV can be increased if the operation cannot be done in time or decreased if you want the experiment to go faster.

Programming exercises

1. Add the while(1) statement at the end of the program in requirement 4 to make the process run forever. Go to System Monitor and operate to end this process.
2. Continue executing the program with while(1); bring this process into the “background” via sending the SIGTSTP signal by pressing **Ctrl + Z**. Use the ps command to determine its PID and use the kill command to end it.
3. Review the process above, when executing the system how many bytes of memory are allocated? Let's edit the program, declare an array of 1 million integers and initialize their values. When executing an editing program, how many bytes of memory does the system allocate? Please explain this change.

CHECKLIST LAB

- ☐ Start Ubuntu.
- ☐ Run Terminal.
- ☐ Edit a text file .txt, .c
- ☐ Compile and Link a .c source code
- ☐ Create a statically linked library.
- ☐ Create dynamic link library.
- ☐ Run a compiled program.
- ☐ Pass arguments via command line.
- ☐ Create child process.
- ☐ Source code structure for parent process and child process.
- ☐ Create a process tree
- ☐ Synchronously terminate the parent/child process.
- ☐ Call to run a command from within the program.